Q: Given this code, answer the following questions:

Python

```
def categorize_temperature(temp, scale):
    if scale == "C":
        if temp < 0:
            return "Freezing"
        elif 0 <= temp <= 30:
            return "Cold to Warm"
        else:
            return "Hot"
    elif scale == "F":
        if temp < 32:
            return "Freezing"
        elif 32 <= temp <= 86:
            return "Cold to Warm"
        else:
            return "Hot"
    else:
        return "Unknown Scale"
```

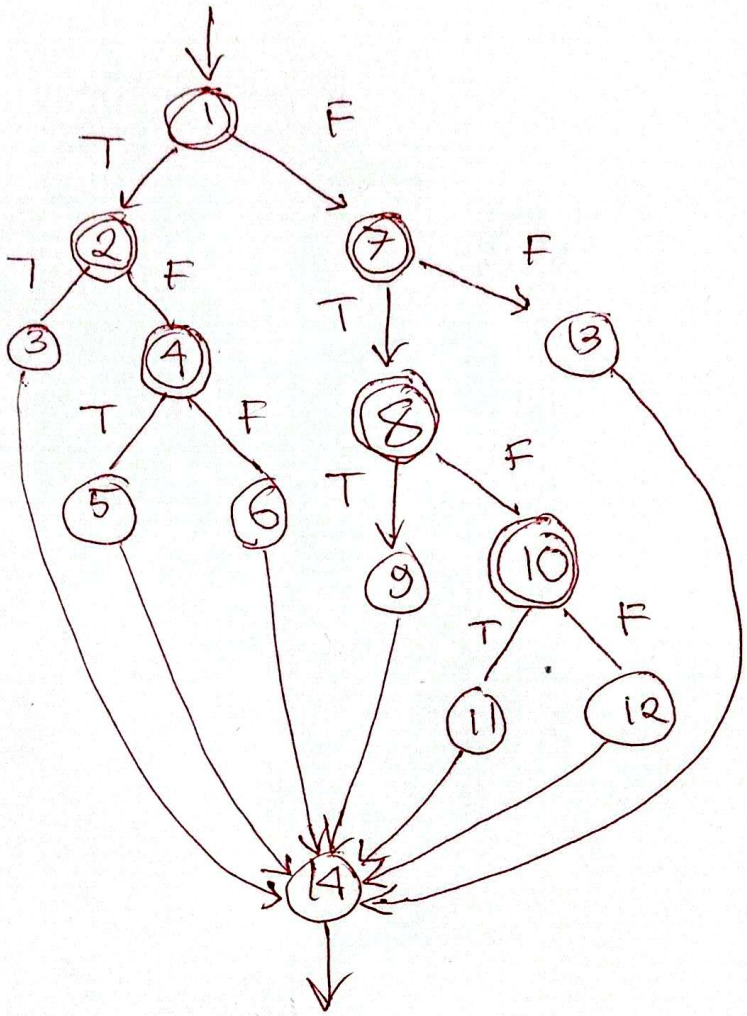
Java

```
public static String categorizeTemperature(double temp, String scale) {
    if (scale.equals("C")) {
        if (temp < 0) {
            return "Freezing";
        } else if (temp >= 0 && temp <= 30) {
            return "Cold to Warm";
        } else {
            return "Hot";
        }
    } else if (scale.equals("F")) {
        if (temp < 32) {
            return "Freezing";
        } else if (temp >= 32 && temp <= 86) {
            return "Cold to Warm";
        } else {
            return "Hot";
        }
    } else {
        return "Unknown Scale";
    }
}
```

1. Draw the CFG for the given method
2. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set
3. Generate test cases for all paths in the Basic Paths Set

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1.



2. $C = P + 1$

$= 6 + 1 = 7$

Basic Paths Set = 1-2-3-14

1-2-4-5-14

1-2-4-6-14

1-7-8-9-14

1-7-8-10-11-14

1-7-8-10-12-14

1-7-13-14

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3.

Test Case:

1-2-3-14:

scale = C temp = -5

1-2-4-5-14

scale = C temp = 5

1-2-4-6-14

scale = C temp = 35

1-7-8-9-14

scale = F temp = 31

1-7-8-10-11-14

scale = F temp = 37

1-7-8-10-12-14

scale = F temp = 100

1-7-13-14

scale = G temp = 20

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Q: Given this code, answer the following questions:

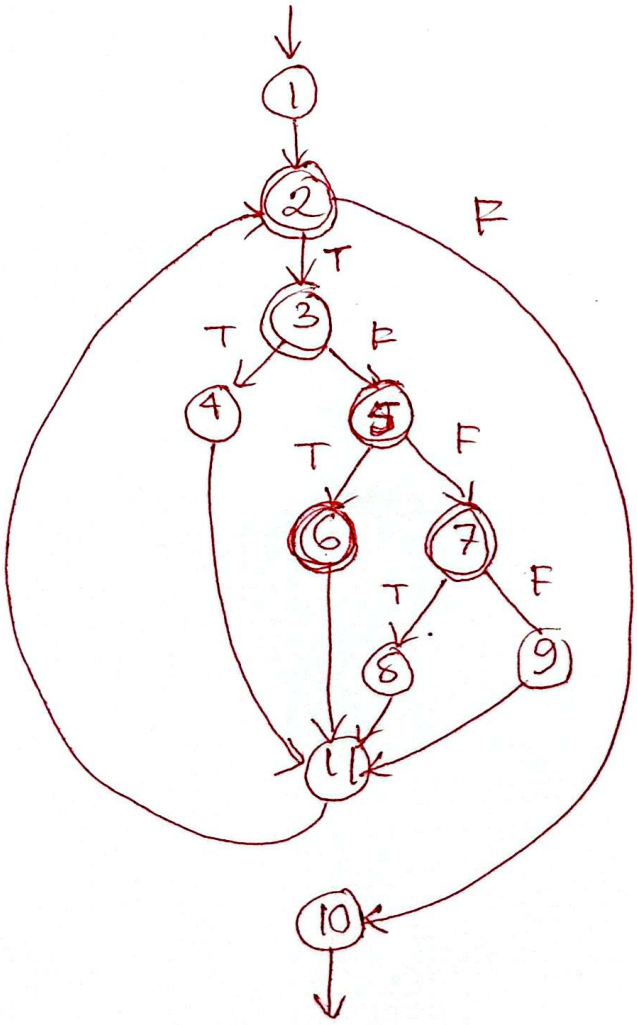
Python

```
def classify_grades(grades):
    result = []
    for grade in grades:
        if grade >= 90:
            result.append("A")
        elif 80 <= grade < 90:
            result.append("B")
        elif 70 <= grade < 80:
            result.append("C")
        else:
            result.append("F")
    return result
```

1. Draw the CFG for the given method
2. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set
3. Generate test cases for all paths in the Basic Paths Set

Answer:

1)



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2) $C = P + 1$

$= 4 + 1 = 5$

Basic Paths Set:

1-2-3-4-11-2 - 3-5-6-11-2 - 35-7-8-11-2

- 3-5-7-9-11-2-10

3) Test Cases:

grades = [92, 85, 75, 20]

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Q: Given this code, answer the following questions:

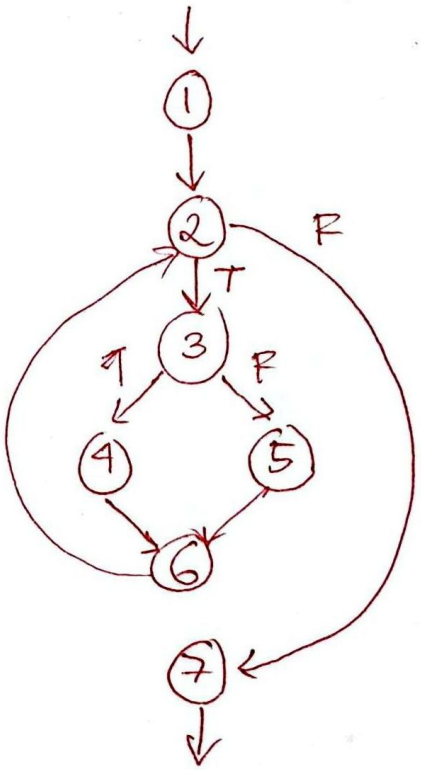
Python

```
def countdown_even_odd(start, step):
 1 result = []
 2 while start > 0:
     3 if start % 2 == 0:
     4     result.append(f"{start} is even")
     5 else:
           result.append(f"{start} is odd")
     6 start -= step
 7 return result
```

1. Draw the CFG for the given method
2. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set
3. Generate test cases for all paths in the Basic Paths Set

Answer:

1)



2) $C = P + 1$

$= 2 + 1 = 3$

Basic Paths Set : 1-2-3-4-6-2-3-5-6-2-7

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3.

Test case:

Start = 2

Step = 1

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Q: Given this code, answer the following questions:

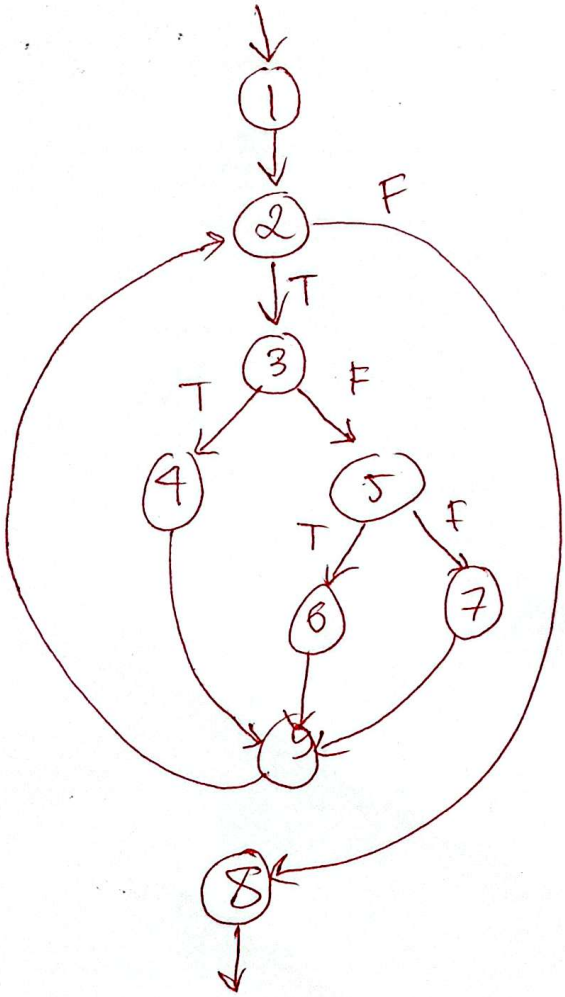
Python

```
def filter_positive_numbers(nums):
    filtered = []
    for num in nums:
        if num < 0:
            continue
        elif num == 0:
            filtered.append("Zero")
        else:
            filtered.append(num)
    return filtered
```

1. Draw the CFG for the given method
2. Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set
3. Generate test cases for all paths in the Basic Paths Set

Answer:

1)



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2) C: P+1

= 3+1 = 4

~~Test Case~~

Basic Paths Set:

1-2-3-4-9-2-3-5-6-9-2-3-5-7-9-2-8

3) Test Cases:

nums = [-1, 0, 3]

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